

INTERNET TECHNOLOGY & WEB PAGE DESIGN

Unit-1: Internet Fundamentals

Comprehensive Study Notes

HPU BCA Semester IV

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1. What is the Internet?

The **Internet** is a global network of interconnected computer networks that use the Internet Protocol Suite (TCP/IP) to communicate with each other.

"The Internet is a worldwide system of computer networks in which users at any one computer can get information from any other computer."

2. History of Internet

Year	Event	Description
1969	ARPANET	First packet-switching network
1972	Email	Electronic mail introduced
1974	TCP/IP	Transmission Control Protocol/Internet Protocol
1983	DNS	Domain Name System introduced
1989	WWW	World Wide Web proposed by Tim Berners-Lee
1991	First Website	The first website went live
1993	Mosaic	First graphical web browser
1995	Internet Explorer	Microsoft's web browser
1998	Google	Google search engine founded
2000s	Social Media	Rise of social media platforms
2010s	Mobile Internet	Smartphones and mobile internet
2020s	IoT	Internet of Things, 5G

3. Major Internet Applications

3.1 Communication Applications

Application	Description	Examples
Electronic Mail (Email)	Sending messages electronically	Gmail, Outlook, Yahoo
World Wide Web	Accessing information via web pages	Chrome, Firefox, Edge
File Transfer Protocol (FTP)	Transferring files	FileZilla, WinSCP
Instant Messaging	Real-time text communication	WhatsApp, Messenger, Telegram
Video Conferencing	Real-time video communication	Zoom, Google Meet, Teams
Search Engines	Finding information online	Google, Bing, Yahoo
Social Media	Social networking	Facebook, Instagram, Twitter
E-Commerce	Online shopping	Amazon, Flipkart

3.2 Categories of Internet Applications

COMMUNICATION	INFORMATION
APPLICATIONS	ACCESS
Email	Search Engines
Instant Messaging	Web Browsers
Video Conferencing	News Groups
VoIP	FTP Archives

4. Basic Network Requirements

Requirement	Description
Hardware	Computers, routers, switches, modems
Software	Operating systems, protocols, applications
Communication Medium	Cables (Ethernet, Fiber) or Wireless (Wi-Fi, Cellular)
Protocols	TCP/IP, HTTP, FTP, SMTP
Addressing	IP Addresses, Domain Names
Security	Firewalls, Encryption, Authentication

4.1 Network Components

Component	Description
Routers	Connect different networks
Switches	Connect devices within a network
Modems	Modulate/demodulate signals
Access Points	Provide wireless connectivity
Network Interface Cards	Connect devices to network

5. Bandwidth

Bandwidth is the maximum amount of data that can be transmitted over a network connection in a given amount of time.

"Bandwidth is the capacity of a communication channel to transmit data measured in bits per second (bps)."

5.1 Bandwidth Units

Unit	Value
bps	bits per second
Kbps	Kilobits per second (1000 bps)
Mbps	Megabits per second (1,000,000 bps)
Gbps	Gigabits per second (1,000,000,000 bps)

6. Electronic Mail (Email)

Email is a method of exchanging digital messages between users.

6.1 Email Components

- **Sender:** Sends the email
- **SMTP Server:** Sends mail to the destination
- **Mail Server:** Stores the email
- **Receiver:** Receives the email

6.2 Email Protocols

Protocol	Purpose	Details
SMTP	Sending emails	Simple Mail Transfer Protocol
POP3	Downloading emails	Post Office Protocol v3
IMAP	Syncing emails across devices	Internet Message Access Protocol
MIME	Multimedia email attachments	Multipurpose Internet Mail Extensions

7. Newsgroups

Newsgroups are discussion forums on the Internet where users can post and read messages.

"Newsgroups are online discussion groups where people can share information and opinions on specific topics."

7.1 Popular Newsgroup Categories

Category	Description
rec.*	Recreation and hobbies
comp.*	Computers and technology
sci.*	Science and research
soc.*	Social issues
talk.*	General discussions
alt.*	Alternative topics

8. FTP Archive

File Transfer Protocol (FTP) is a protocol for transferring files between computers.

8.1 FTP Features

- File upload and download
- Directory navigation
- File management (delete, rename, move)
- Anonymous access

9. Real-Time Activities

Real-Time Activities allow users to interact instantly with others over the Internet.

Activity	Description	Examples
Instant Messaging	Real-time text chat	WhatsApp, Messenger
Video Conferencing	Real-time video chat	Zoom, Teams
Online Gaming	Multiplayer games	Fortnite, Call of Duty
Live Streaming	Real-time video broadcast	YouTube Live, Twitch
VoIP	Voice over IP	Skype, WhatsApp Calls

10. Video and Audio

Service	Description
Streaming Video	Real-time video playback (YouTube, Netflix)
Streaming Audio	Real-time audio playback (Spotify, Apple Music)
Podcasts	Digital audio episodes
Webcasts	Live or recorded broadcasts

11. Search Engines

Search Engines are tools that help users find information on the Internet.

11.1 Popular Search Engines

- Google
- Bing
- Yahoo
- DuckDuckGo

Questions

Short Answer Questions (2-5 Marks)

1. What is the Internet?
2. What is the history of the Internet?
3. What are the applications of the Internet?
4. What is bandwidth?
5. What is email? What are its protocols?
6. What is FTP?
7. What is a newsgroup?
8. What is a search engine?
9. What are real-time activities on the Internet?
10. What are the network requirements?

Long Answer Questions (10 Marks)

1. Explain the evolution of the Internet.
2. Explain the applications of the Internet.
3. Explain email and its protocols.
4. Explain FTP and its uses.
5. Explain search engines and their working.

Objective Questions (1 Mark)

1. The Internet was originally called:
(a) ARPANET (b) INTERNET (c) NSFNET (d) None
2. WWW was proposed by:
(a) Tim Berners-Lee (b) Bill Gates (c) Steve Jobs (d) None
3. SMTP is used for:
(a) Sending emails (b) Receiving emails (c) Both (d) None
4. POP is used for:
(a) Sending emails (b) Receiving emails (c) Both (d) None

5. FTP stands for:

- (a) File Transfer Protocol (b) Fast Transfer Protocol
- (c) File Transmission Protocol (d) None